For chief data officers and data platform leaders

Why querying enterprise data at scale should not cost a fortune.

The importance of being able to query data at scale, at speed and at a low, predictable cost, and why today's compute economics stop financial institutions from asking the questions regulators expect them to answer.

1TB+
Interactive query target

10s
Euro instant payment window

4h
DORA incident notification

€0
Marginal cost of a query on QBricks

Executive summary

Financial institutions have never held more data, and have never been more reluctant to query it. Modern data management solutions encourage you to put your data in their platform and then charge you compute to query it: storage is cheap, every question is metered, so every full-population screen, every multi-year link analysis and every supervisory replay carries a visible, variable bill. The rational response (sample, pre-aggregate, shorten the look-back) is precisely the behaviour that instant payments, AMLR, DORA, T+1 and the AI Act now penalise.

This paper sets out why querying 1 TB and more, in seconds, is an operating condition for modern financial services rather than a performance nicety; lists the business requirements that depend on it; explains how the economics of current data platforms turn those requirements into budget negotiations; and describes how QBricks delivers querying at scale for a fixed, containerised fee, so the marginal cost of a question is zero.

When asking is free, institutions ask. That is the whole argument.

01 · The obligation

Every obligation ends in a very large query.

Regulation is written in terms of outcomes (detect, notify, settle, reimburse) but each outcome is produced by a question asked of a dataset. A sanctions delta must be screened against the full customer base, immediately, under the Instant Payments Regulation. A DORA classification must count affected clients and transaction value across every channel inside four hours. An AMLA request for a network of related parties reaches back five to seven years. A T+1 settlement break has to be traced through positions, instructions and reference data before the market opens. A credit model under the AI Act's Annex III must show that its training data was representative, and keep proving it in operation.

In a mid-sized institution these datasets are measured in terabytes; in a universal bank, in petabytes. The questions arrive without warning (a list change, an outage, an inspector's email) and the answer is needed in seconds or minutes, not after a weekend job.

The ability to query 1 TB and more, fast, is therefore the operating condition under which every control in the modern rulebook can run at all. An institution that cannot afford to ask the question has not implemented the control; it has documented the intention to.

02 · The requirement

What the business actually needs.

Six recurring requirements across risk, compliance, fraud, operations and data science depend on scale and speed together. None of them can be met by a sample or by a shortened window.

Six requirements that need scale and speed together

Requirement	Why it needs scale and speed	Driver
Full-population screening	Every list change re-screens every customer and beneficial owner, immediately: not a sample, not a delta of recent activity.	IPR Art. 5d · AMLR
Network and link analysis	Mule accounts, layered ownership and shared devices only surface when years of transactions are joined across entities in one pass.	AMLR · PSR · APP
Incident blast-radius counts	Clients, counterparties and value affected must be computed across all channels inside the four-hour window.	DORA Art. 19
Intraday risk and liquidity	Positions, collateral and funding re-aggregated several times on T+0, on demand and under stress.	T+1 · BCBS 239
Supervisory and audit replay	An on-site inspection asks for a given number to be reproduced from source, across the full retention period, while the inspector waits.	ECB RDARR · AMLA
Model training and validation	Representative, bias-tested datasets and post-market monitoring require repeated full-history scans, not one-off extracts.	AI Act Art. 9 and 10

Beyond compliance, the same capability drives the commercial agenda. Customer-level profitability, real-time pricing, churn and next-best-action models all improve with the depth of history they can see. The data office that can afford to scan five years answers a different class of question from the one that can afford ninety days.

03 · The rationing

Why current platforms make institutions ration the question.

Modern data management solutions encourage you to put your data in their platform, then charge you compute to query it. The commercial model rewards the vendor every time you ask a question of your own data, and the more valuable the question, the more it costs to ask. There has to be a better, more cost-effective way.

Cloud data platforms solved storage; they did not solve the economics of compute. Query cost scales with data scanned, concurrency and duration, and is billed per second, per credit or per terabyte processed. The effect on behaviour is predictable.

Queries are rationed. Teams sample, pre-aggregate or restrict look-back windows to protect budgets: exactly the shortcuts that break full-population screening and lineage replay.

Cost is unpredictable. A sanctions list change, a market event or a supervisory request produces a compute spike that nobody budgeted for, so the controls that matter most are the ones finance pushes back on.

Duplication multiplies the bill. Each use case builds its own pipeline and its own copy of the data, and every copy is re-scanned by every job that touches it.

Peak capacity is paid for permanently. Warehouses are sized for the worst day, or auto-scale into it with no ceiling. Either way the CDO owns a variable cost line that the business reads as a reason to ask fewer questions.

Governance gets charged twice. Quality checks, lineage capture and reconciliation are themselves compute jobs. Platforms that meter compute make good governance more expensive than bad governance.

The governance paradox. The institution holds the data that would answer the regulator's question, but has built incentives not to ask it.

When every query has a price, the first casualty is the query that matters most.

04 · The fixed fee

A fixed, containerised fee for querying at scale.

QBricks removes the variable from the equation. Its query and ingestion engine runs as a container inside your environment, including on local compute where residency demands it, for a fixed, containerised fee. Capacity is sized once, to the estate, and the price does not move with the number of queries run, the terabytes scanned or the day of the month.

Scan once, govern once

Data lands incrementally under an ODCS data contract, so there are no duplicate copies and no reprocessing of history on every run. Full-population queries read a single governed estate.

Terabytes in seconds, inside the container

The engine is built for 1 TB+ interactive queries such as screening, link analysis and replay, without spinning up and paying for a transient cluster each time.

A number the CFO can sign

One fee per container per year. Compliance, risk, fraud and data science query the same estate as often as they need; the invoice is the same in a crisis month as in a quiet one.

Your platform, your security

The container inherits the host platform's identity, encryption and access controls; data never leaves your tenancy, and agents remain isolated from the lake.

Governance and performance are not traded off against each other. Because every record is checked against its contract on the way in, the estate being queried is already clean, deduplicated and lineaged. There is no reconciliation layer to pay for before the analytical query can start, and every result carries its before-and-after trail for the auditor.

The behavioural change is the point. When the marginal cost of a question is zero, the sanctions team screens everyone every time, the fraud team joins five years instead of ninety days, and the data office answers the supervisor on the spot. Querying at scale stops being a budget negotiation and becomes the default, which is exactly what the legislation assumes it already is.

05 · Metered versus fixed

Metered versus fixed: what changes.

Dimension	Metered compute	QBricks fixed container
Cost of one more query	Positive and visible; scales with bytes scanned	**Zero**
Budget behaviour	Monthly variance; spikes on the days that matter	**Flat annual line; a crisis month costs the same**
Look-back window	Shortened to control spend	**Full retention period, every time**
Data copies	One per pipeline and team; each re-scanned	**One governed estate, scanned once**
Governance overhead	Extra jobs, extra compute, often deferred	**Applied at ingestion, included in the fee**
Who decides whether to ask	Finance	**The analyst**

06 · Next steps

What to do next.

01 **Measure the rationing.** List the controls and analyses that currently run on samples, shortened windows or pre-aggregates purely for cost reasons. Each is a latent finding.

02 **Price a full-population query today.** Take one, such as a complete sanctions re-screen or a five-year link analysis, and establish its true compute cost and elapsed time on the current platform.

03 **Count the copies.** Inventory how many times the same regulated domain exists across pipelines and teams, and what each copy costs to scan.

04 **Put contracts in front of ingestion.** Govern once, on the way in, so the queryable estate is the audited estate.

05 **Run the same query on a QBricks container.** Compare elapsed time, lineage output and the invoice, then ask whether the business would behave differently if that were the permanent price.

No more data pipelines.

Bring one terabyte-scale question to a QBricks briefing and see it answered on a governed estate, at a price that does not change.

Request a briefing

Regulatory references

1. Regulation (EU) 2024/886 (Instant Payments Regulation), Art. 5d: at-least-daily customer sanctions screening, immediate on list change.
2. Regulation (EU) 2024/1624 (AMLR), applying 10 July 2027: ongoing monitoring, five-to-seven-year retention, Art. 75 information-sharing partnerships.
3. Regulation (EU) 2022/2554 (DORA) and Delegated Regulation (EU) 2025/301: four-hour initial notification of major ICT incidents.
4. CSDR Art. 5 amendment and UK Statutory Instrument: T+1 settlement from 11 October 2027.
5. BCBS 239 and ECB Guide on effective risk data aggregation and risk reporting (May 2024): on-demand aggregation with lineage.
6. Regulation (EU) 2024/1689 (AI Act), as amended by Regulation (EU) 2026/1744: Annex III high-risk obligations from 2 December 2027; Art. 9 and 10 data governance.

Companion paper: Regulators feel the need, the need for speed (QBricks, October 2026) sets out each instrument in detail. Product performance and pricing statements describe the QBricks containerised model; confirm figures with Infinium Consulting for your estate. This paper is not legal advice.

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